IEOR E4004 Optimization Models and Methods, Section 001

Milestone 2: Problem Definition & Data Exploration

Team 23（L） October 9, 2026

Title: Optimization for a Northeast Corridor Natural Gas Merchant

Problem Description:
A natural gas merchant based in Pennsylvania, modeled on BP's U.S. gas marketing business, buys Marcellus/Utica gas and delivers it to contracted wholesale customers in Pennsylvania, New York, and New Jersey. Higher winter demand can raise gas prices [1], and pipeline constraints into New York and the broader Northeast can become more severe during winter peaks[2]. The merchant must decide when to buy, how much to store, and how to route gas so it meets its contracts at minimum cost.

Research Question:
Over one storage year (April–March), what monthly purchase, pipeline shipment, and storage schedule minimizes the merchant's total cost of meeting its contracted demand in PA, NY, and NJ?

Method:
The project will formulate a multi-period linear programming model. The decision variables determine monthly natural gas supply, transportation flows, and storage inventory. The objective is to minimize purchasing, transportation, and storage costs subject to regional demand, pipeline capacity, storage capacity, and flow-balance constraints. To keep the model manageable, we will use a few pipeline routes and one PA storage site, with limits on injections and withdrawals. Ending inventory in March will equal starting inventory.

Data:
The proposed baseline is April 2025–March 2026. Monthly natural gas demand, prices, storage data, and pipeline capacity information will primarily come from the U.S. Energy Information Administration (EIA). Tariff information will be obtained from the Federal Energy Regulatory Commission (FERC), with simplified transportation-cost assumptions used where detailed tariff data are not practical. EIA currently provides state-level monthly consumption/pricing/storage series and pipeline capacity data, while FERC provides tariff-related information.

Data needed	**Source**	**Working link**	**What it provides**	Availability	What we still need
Gas prices	EIA	Henry Hub prices	Monthly price at Henry Hub ($/MMBtu).	Available	We use Henry Hub as a starting point. Choose a PA buying price or estimate the price difference. Prices at utility delivery points are for comparison only.
Storage capacity	EIA	Storage capacity	Usable gas storage space by state and site.	Available	Choose a PA storage site and assume how much capacity the merchant can use. Set daily limits on gas going in and out.
Inventory	EIA	Weekly inventory	Weekly usable gas in storage by region (Bcf).	Available	Regional data show seasonal patterns. Set starting gas within the storage limit. End March with the same amount.

Pipeline capacity	EIA	Pipeline data	Daily pipeline capacity between states, by direction (MMcf/day).	Available	Assume the merchant can use a set share of each route. Check that the routes connect and can carry gas between the chosen points. Actual contract rights may be private.
Demand	EIA	Consumption data	Monthly gas delivered to customers, by state and customer type (MMcf).	Some gaps	Assume the merchant supplies a set share of PA/NY/NJ demand. Estimate five missing monthly totals using earlier industrial data for the same month. Test how the estimates affect results.
Transportation cost	FERC	eTariffViewer	Fees and rules for moving gas through pipelines.	To be chosen	Choose a route and service. Separate fees for reserving capacity from fees for moving gas, and include gas used as fuel. Check rate dates and estimate missing costs.
Storage cost	FERC	eTariffViewer	Fees and rules for using gas storage.	To be chosen	Choose a storage service. Check fees for reserving space and adding or removing gas, daily limits, seasonal rules, and gas lost during storage.

Extension:

Shadow prices of capacity. We will use LP duality to estimate the marginal value of additional pipeline and storage capacity. Interpreted in the units of each capacity constraint, these values will help identify which capacity contracts warrant further consideration.

Price and demand Monte Carlo. We will simulate winter price and monthly demand scenarios, including higher NY/NJ demand. We will evaluate the base plan's cost and feasibility, identify scenarios where it cannot meet commitments, and specify any permitted adjustments before measuring adaptation costs. The monthly model addresses seasonal planning rather than daily cold-snap reliability.

References:

[1] U.S. Energy Information Administration, *Short-Term Energy Outlook*, Oct. 2026, p. 9.
[2] New York Independent System Operator, *2023 State of the Market Report*, 2024, p. 95